ASSIGNMENT 1 — Q4

Academic Performance Evaluator

JavaScript solution

```
// Q4: Academic Performance Evaluator
"use strict";

const marks = [88, 91, 84, 79, 90]; // Replace with the student's marks.
const totalMarks = marks.reduce((sum, mark) => sum + mark, 0);
const average = totalMarks / marks.length;
const percentage = (totalMarks / (marks.length * 100)) * 100;
const hasFailedSubject = marks.some(mark => mark < 35);

let result;
if (hasFailedSubject || percentage < 50) {
    result = "Detained";
} else if (percentage >= 85 && average >= 85) {
    result = "Promoted with Distinction";
} else if (percentage >= 50 && percentage < 85) {
    result = "Promoted";
} else {
    result = "Detained";
}

console.log(`Average: ${average.toFixed(2)}`);
console.log(`Percentage: ${percentage.toFixed(2)}%`);
console.log(result);
```

Logic: Validates every subject first; any mark below 35 results in Detained. The sample marks can be replaced with actual marks.

How to test: Save this code as a .js file and run it with Node.js using `node filename.js`, or paste it into the browser console.